Math 1200 Review Session 1



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Office hours: Friday TBD

Why Multivariable Calculus?

Meet your Neighbors

- Name
- Year
- House
- Intended major
- Either: What's something you hope to accomplish in the next 4 years

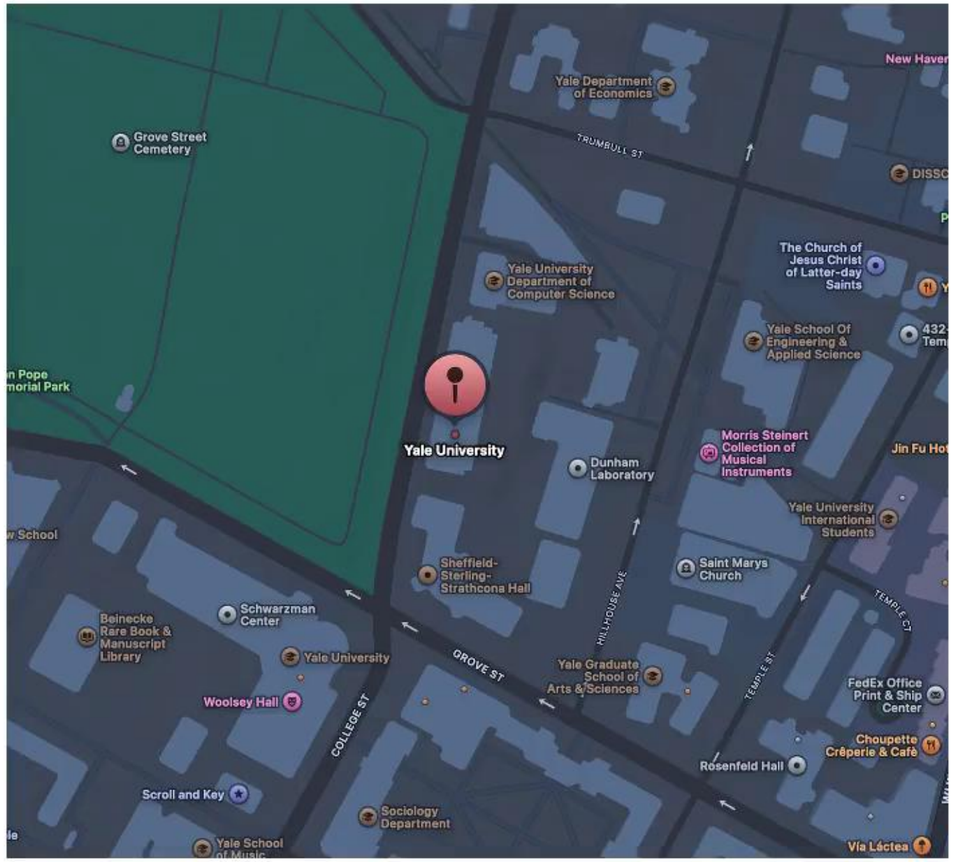
  OR If you had to choose a point on the 2d coordinate plane what point would you choose?

Meet your Neighbors

$(41.31246, -72.92517)$

- Name
- Year
- House
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- Either: What's something you hope to accomplish in the next 4 years

  OR If you had to choose a point on the 2d coordinate plane what point would you choose?



Today's Topics

- Algebra: comfort with substitution
- Trigonometry: no need to memorize
- Parametric Equations: putting it all together
- Tips and Tricks

Pre-class questions

1) Solve the system of equations:

$$e^x - 3y = 7$$
$$e^x + 2y = -1$$

2) $\sin(\theta) = 2/\sqrt{5}$. What is $\tan(\theta)$?

3) Sketch the path of a particle following the parametric equations:

$$(x(t), y(t)) = (3\cos(2t),\ -4\sin(2t))$$

Indicate where the particle starts and which direction it is moving.

Algebra

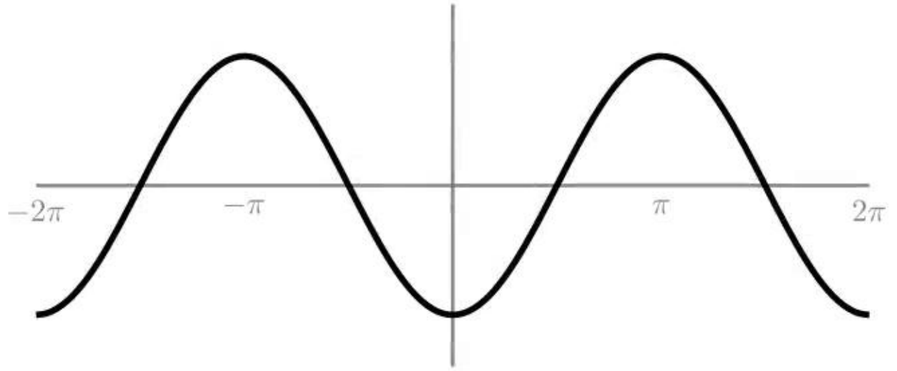
Solve for y as a function of x:

$$x = \tfrac{1}{2}e^{-3t} \qquad y = e^{-6t} + 2e^{-3t} - 8$$

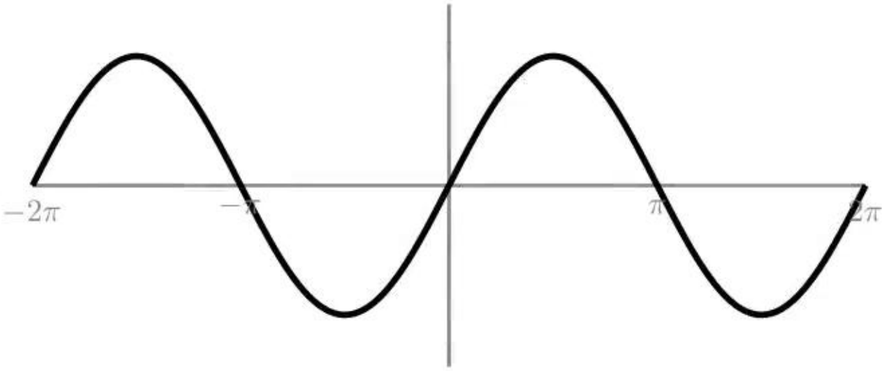
Trigonometry

Which of the following graphs is $y = \sin(x)$?

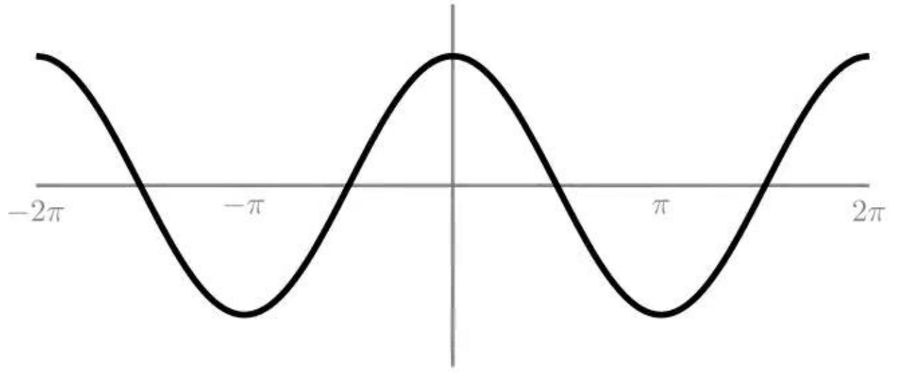
A)



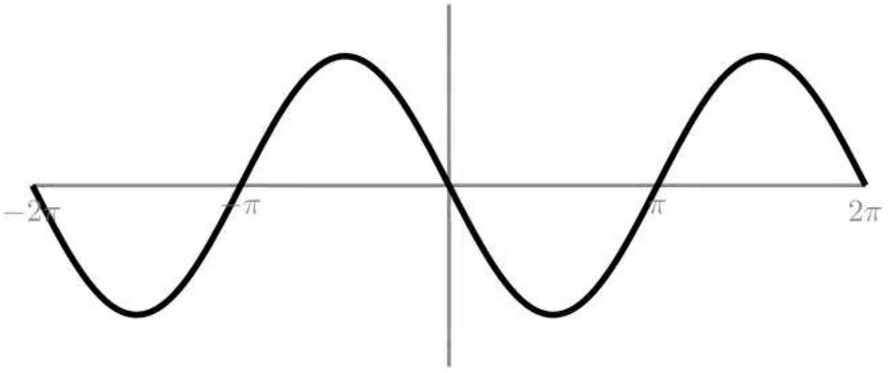
B)



C)

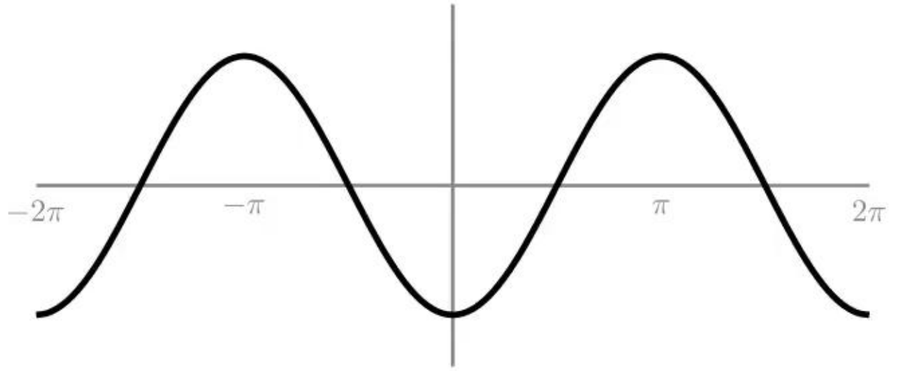


D)

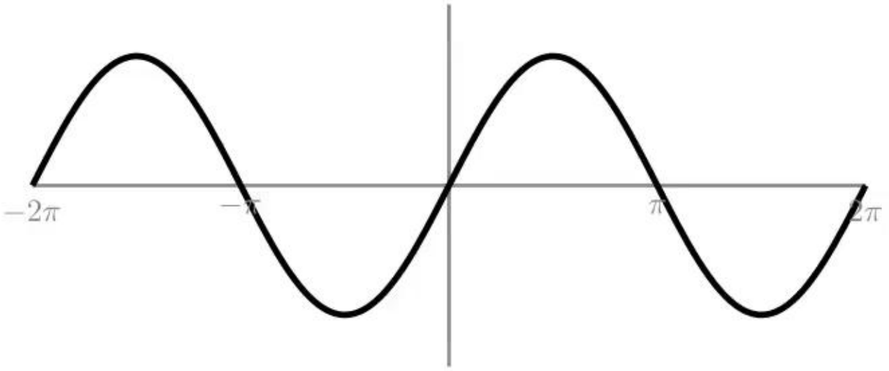


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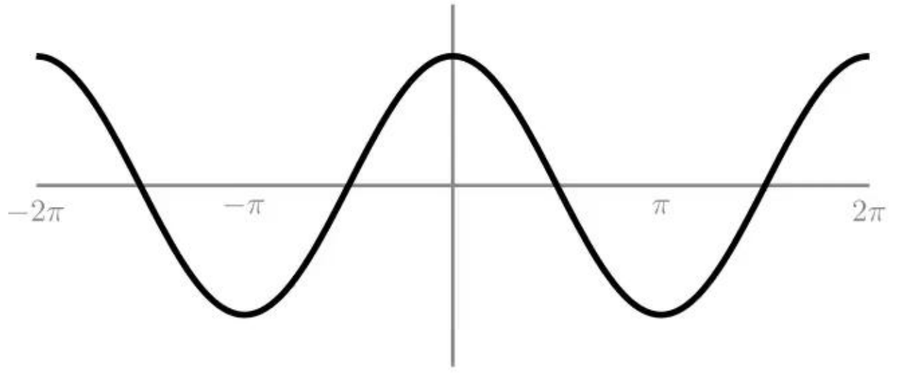
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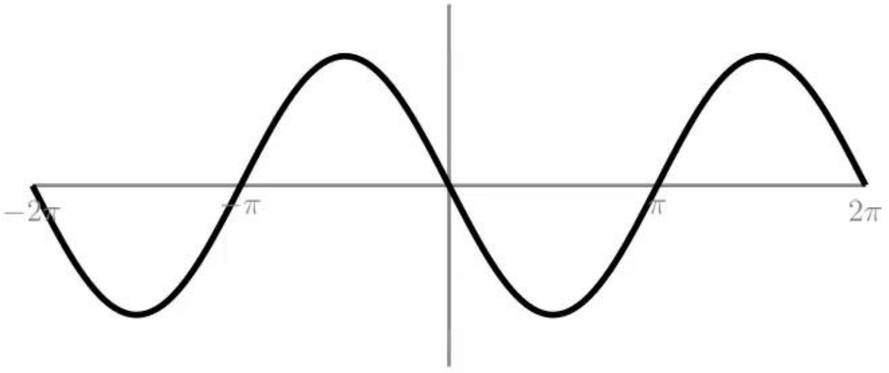
B)



C)

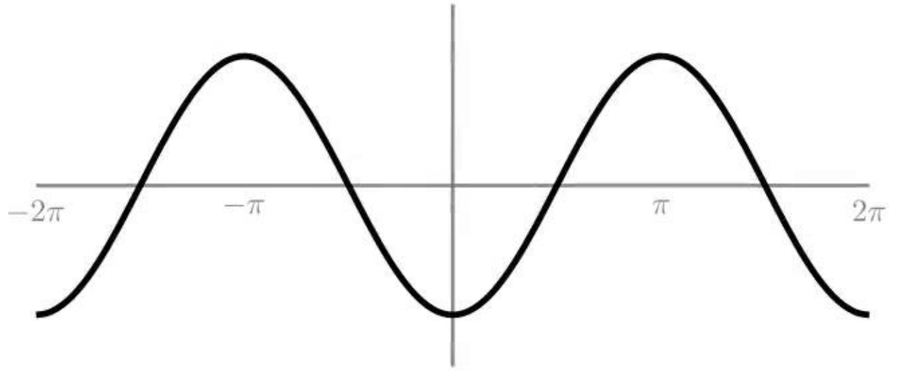


D)

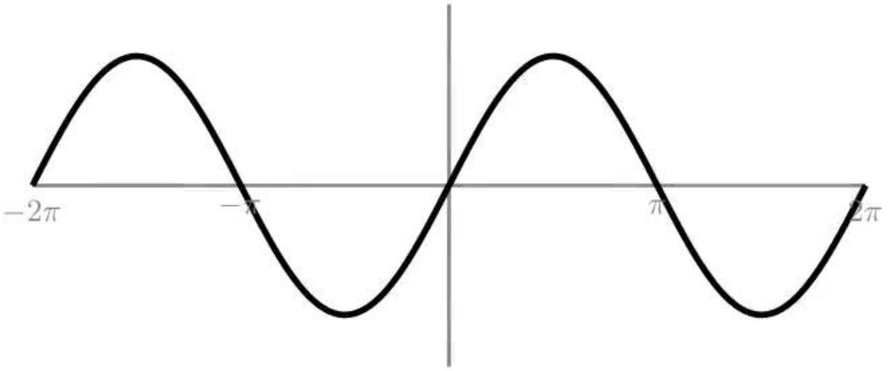


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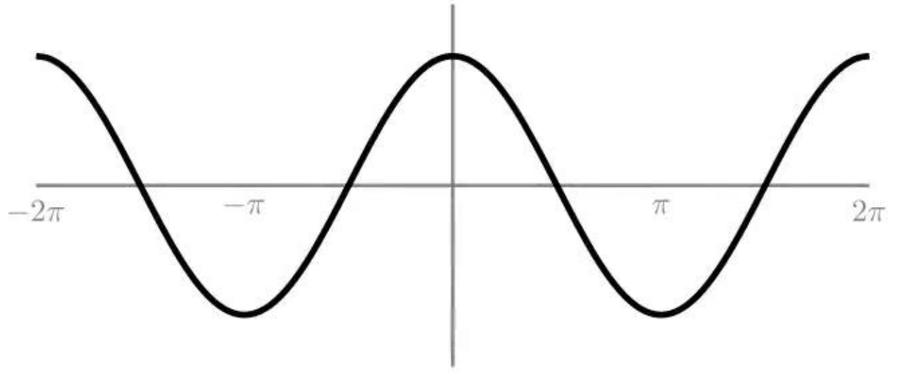
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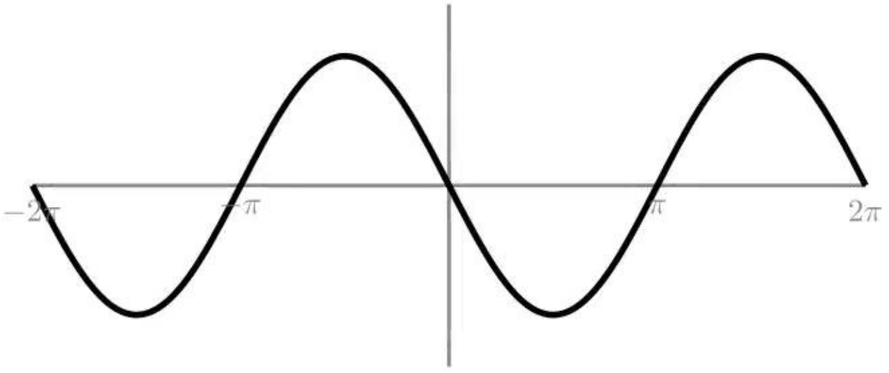
B)



C)

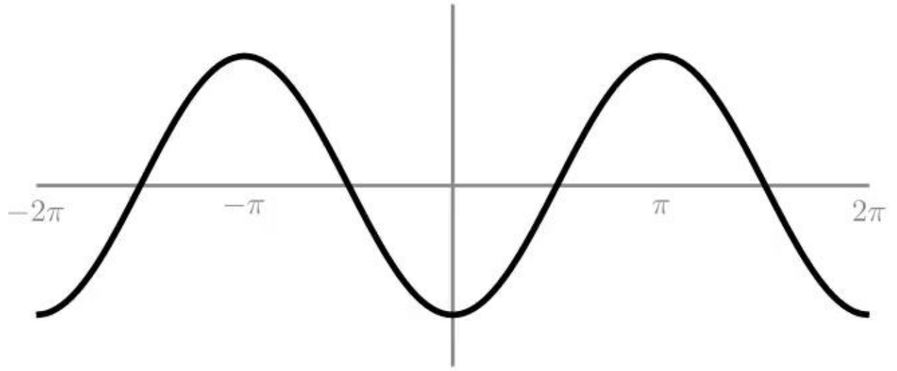


D)

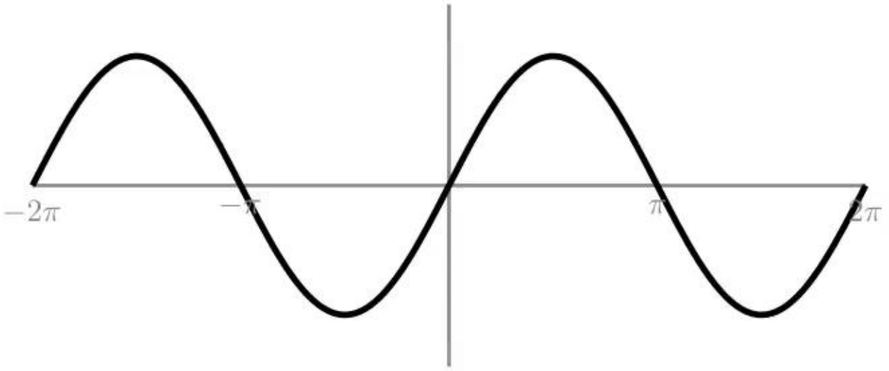


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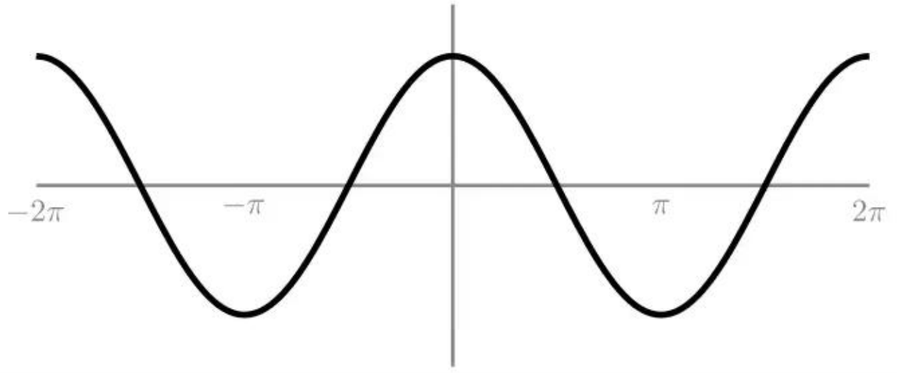
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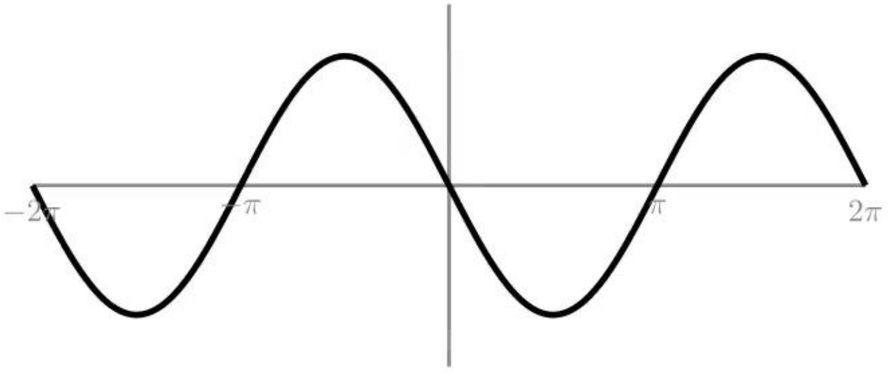
B)



C)



D)



What is $\sin(x)$?

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SOHCAHTOA

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~~**SOHCAHTOA**~~

What is $\sin(x)$?

~~SOHCAHTOA~~



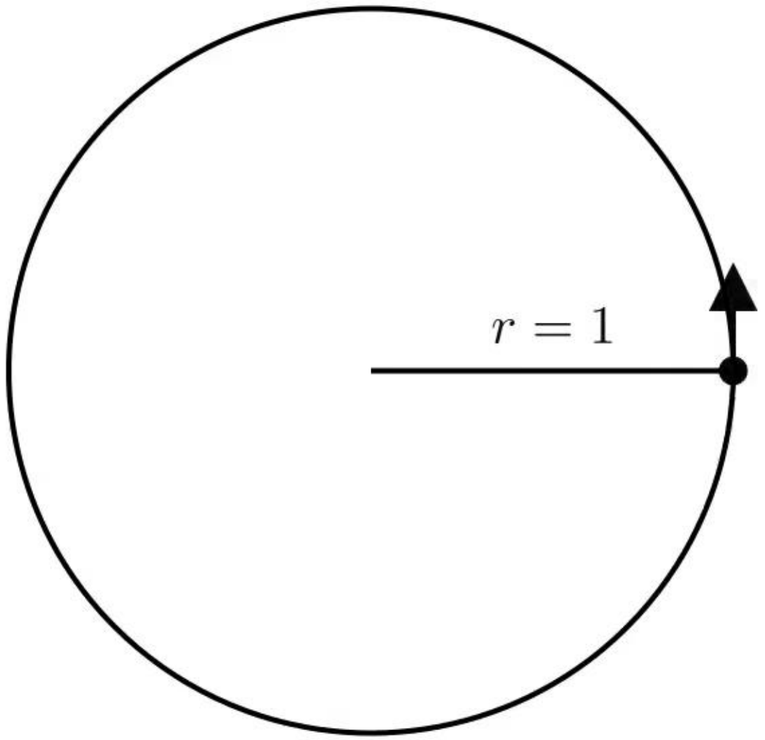
Veritasium, "Inside The Navy's Indoor Ocean," YouTube.

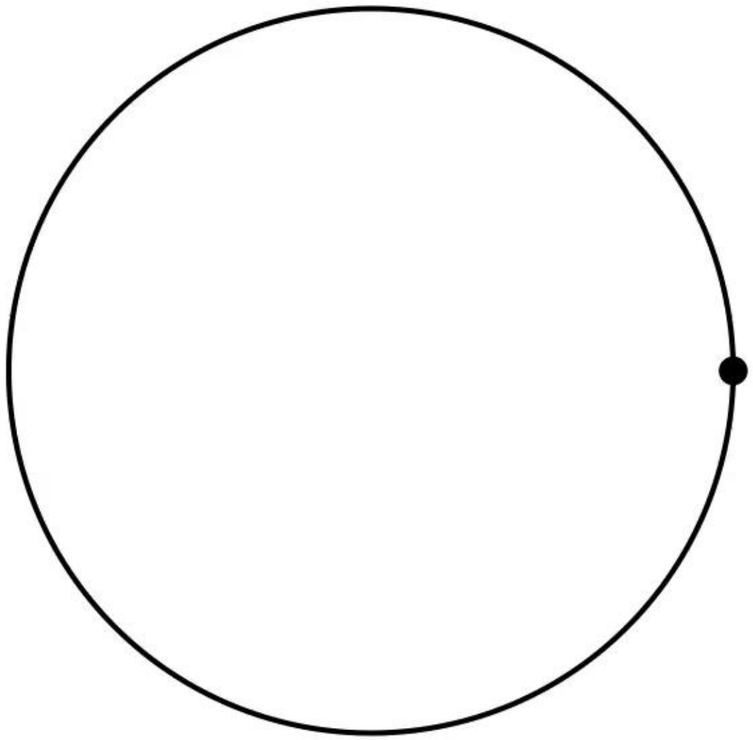
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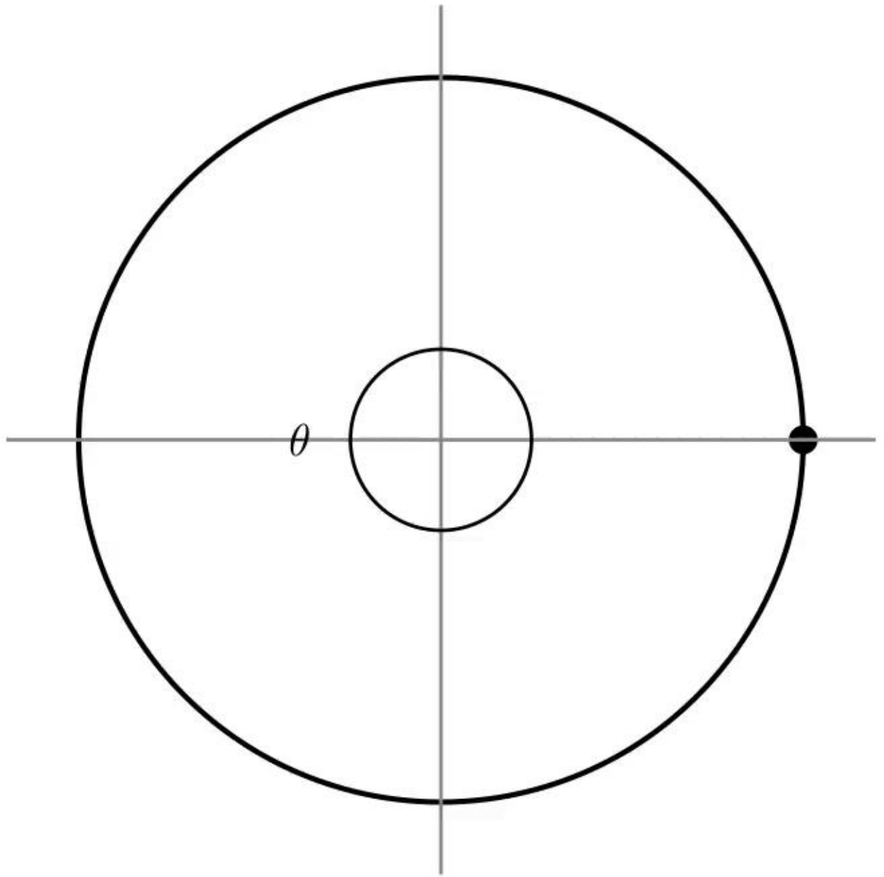
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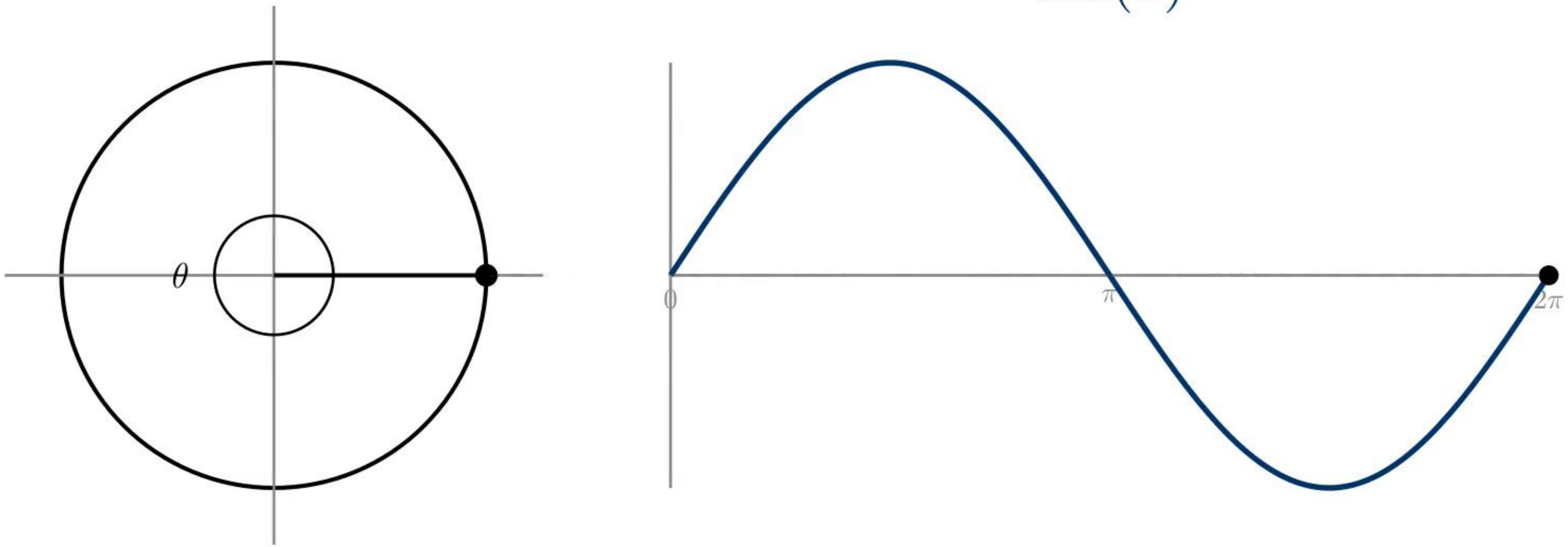


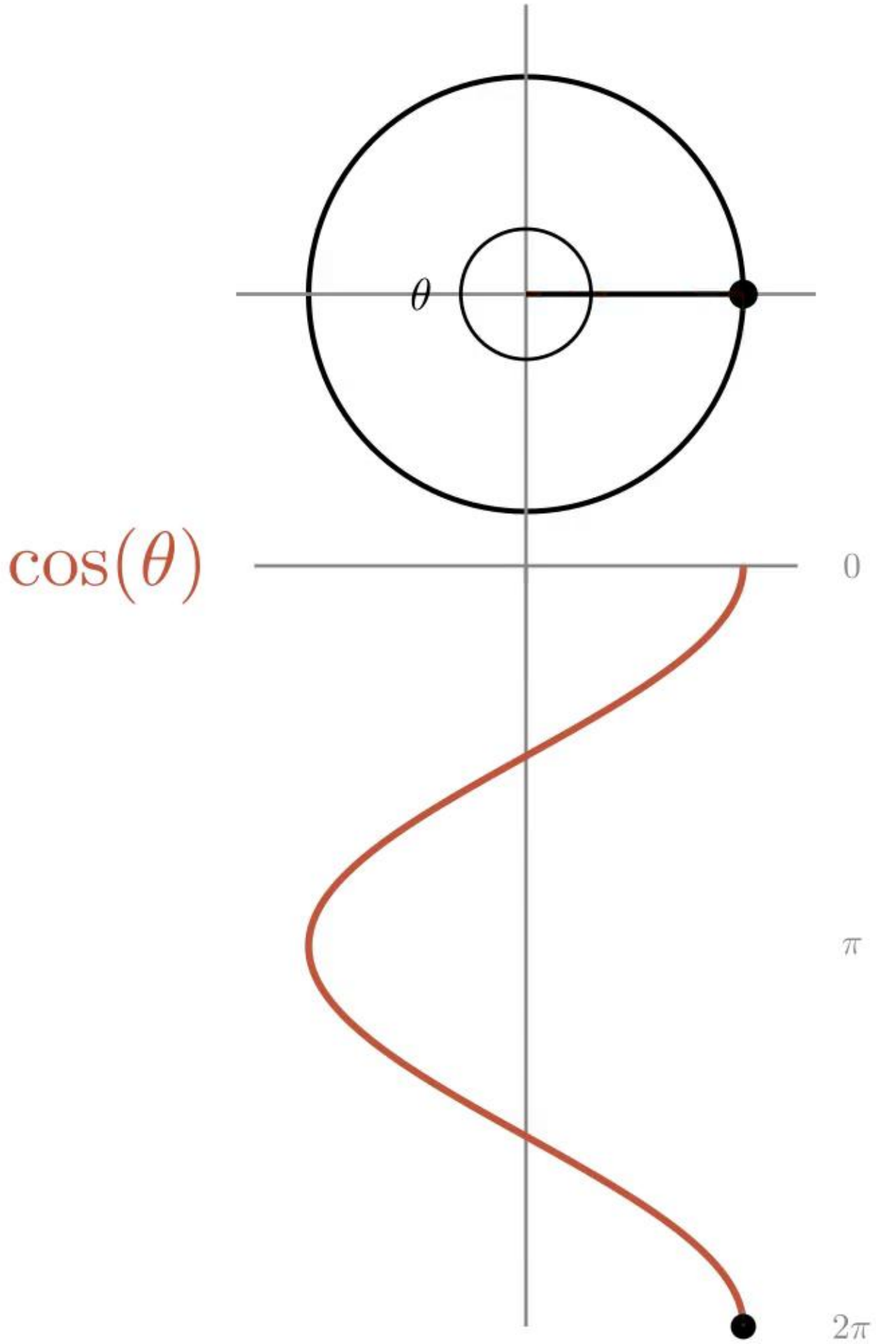
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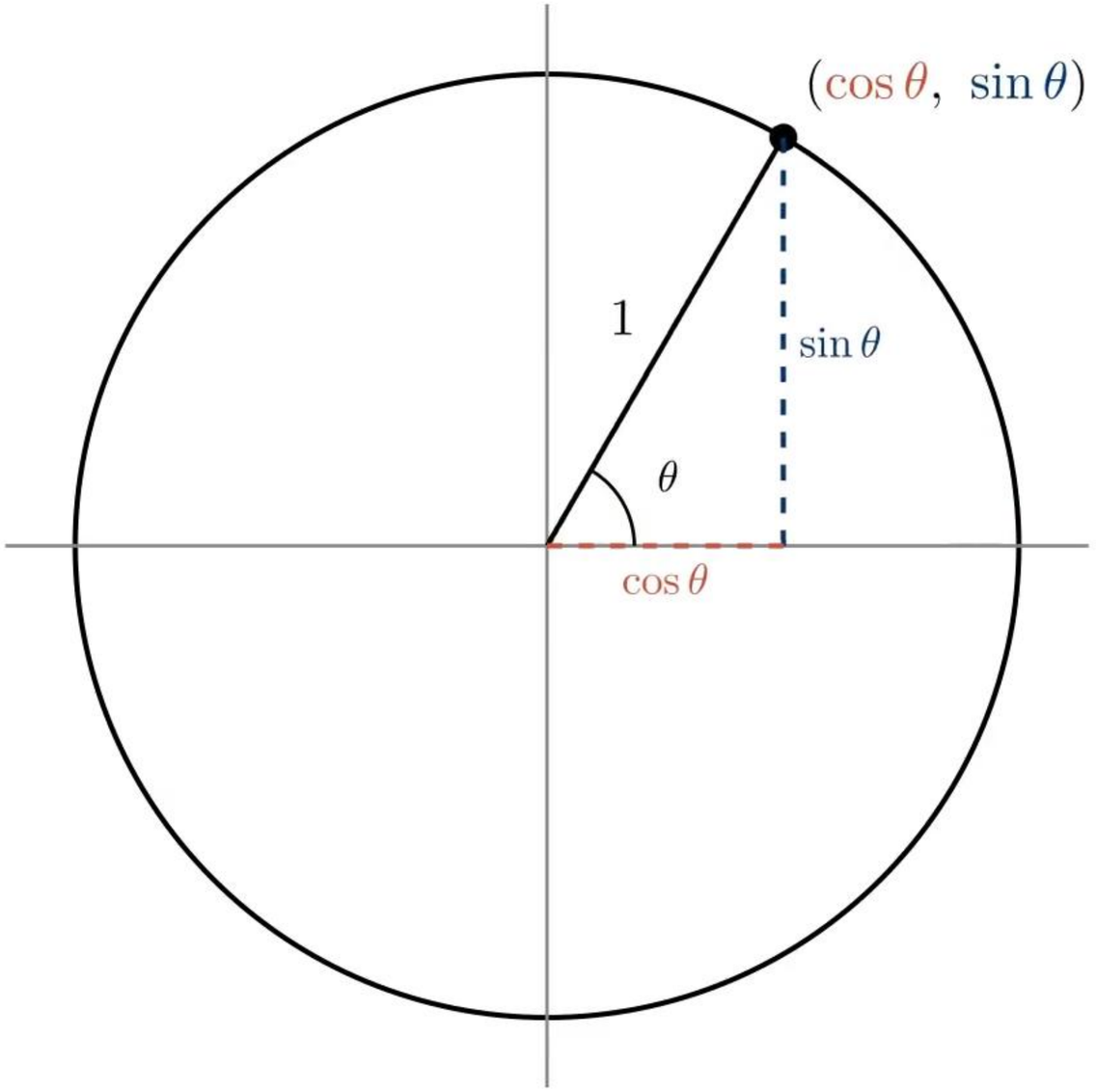

$r = 1$

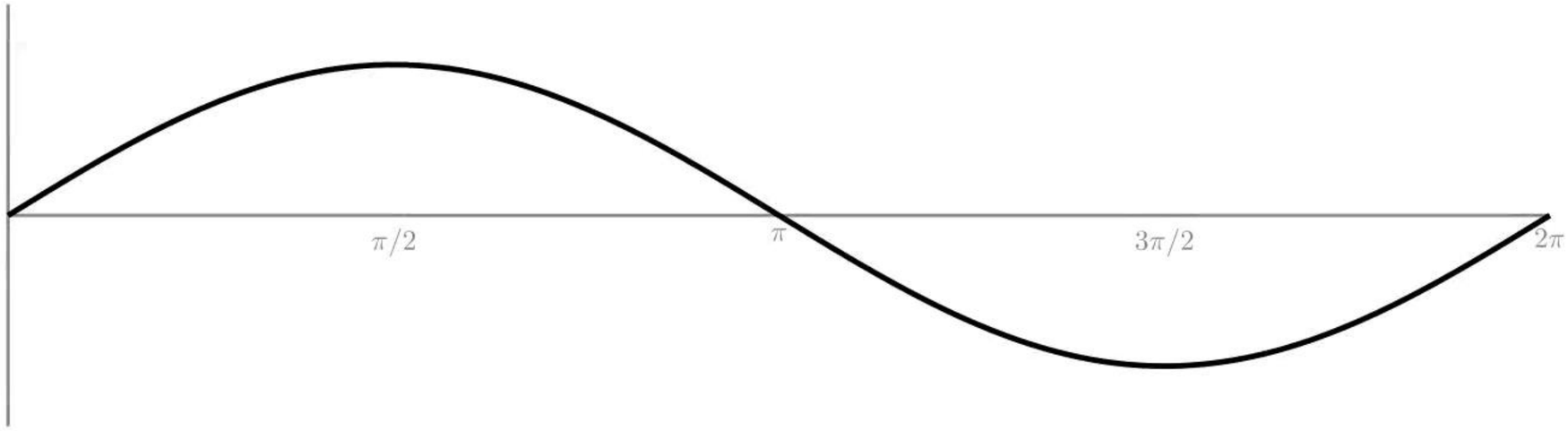



θ

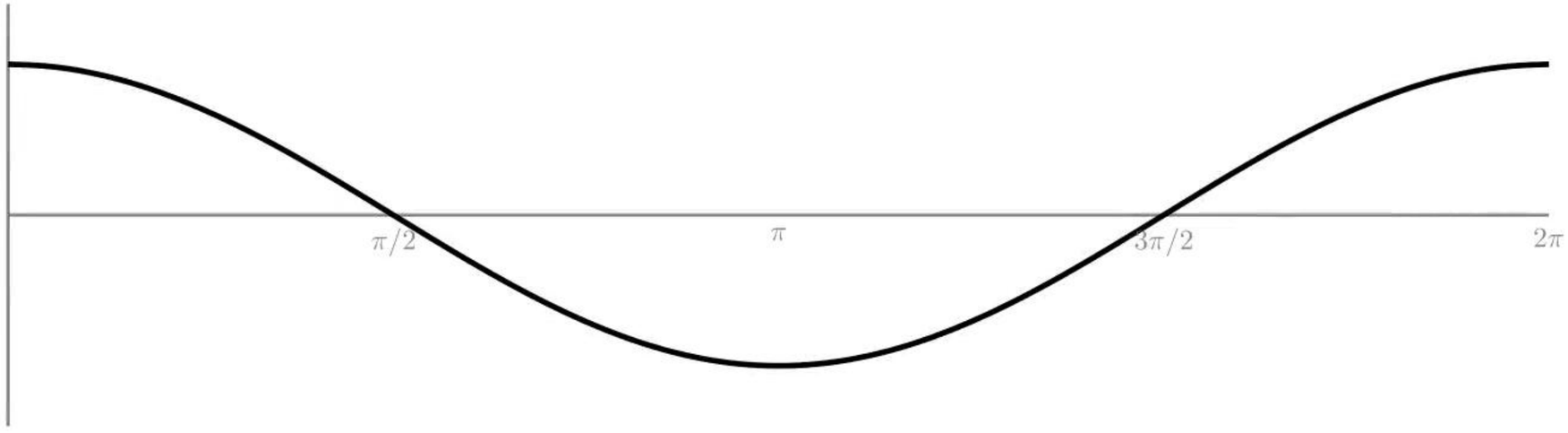

$\sin(\theta)$
θ
0
π
2π


θ
$\cos(\theta)$
0
π
2π


$(\cos\theta,\ \sin\theta)$
1
$\sin\theta$
θ
$\cos\theta$



$y = \sin(x)$

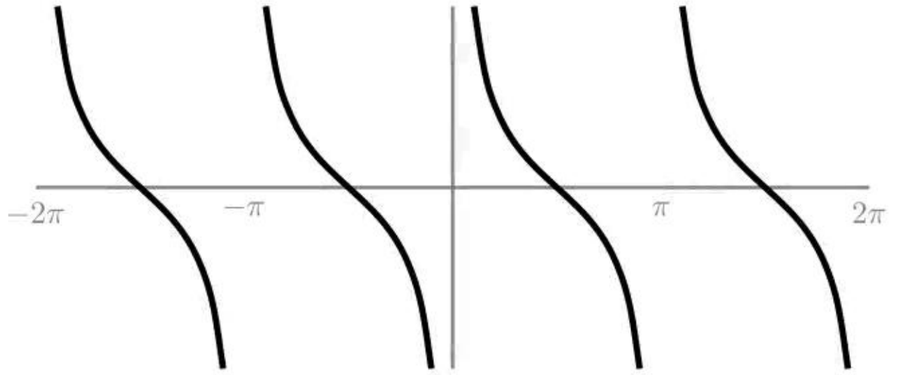


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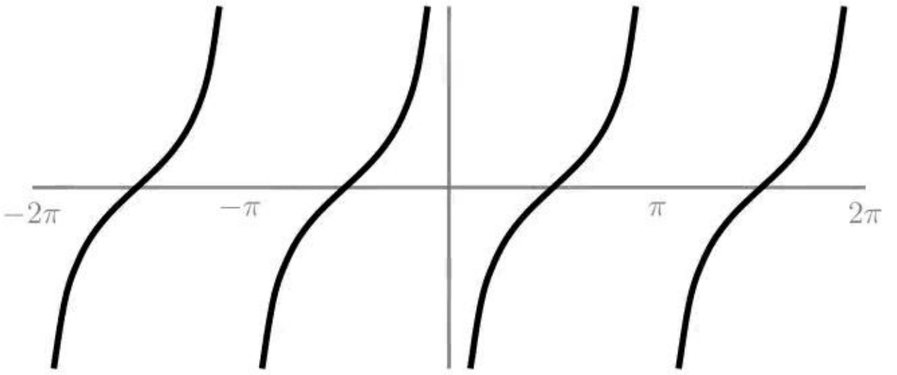
WRITE THESE DOWN

Which of the following graphs is $y = \tan(x)$?

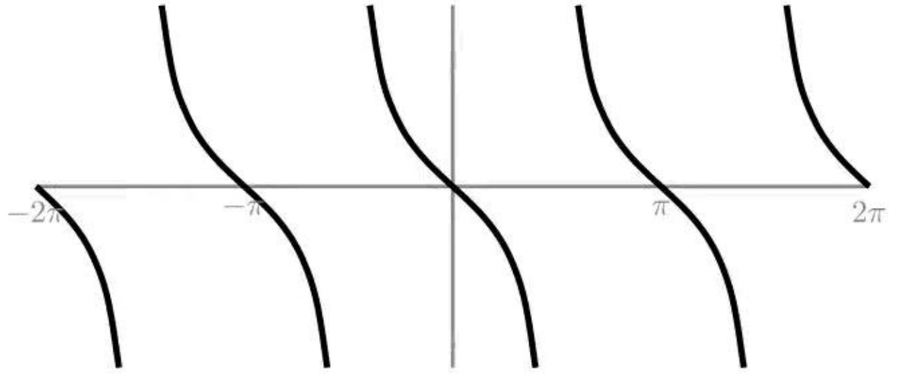
A)



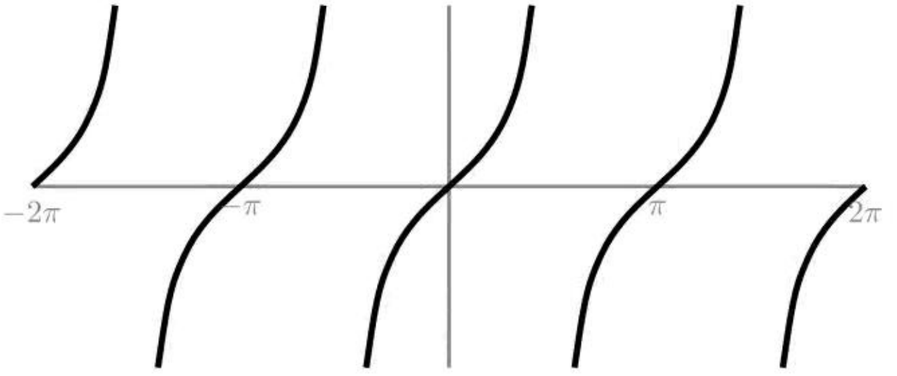
B)



C)

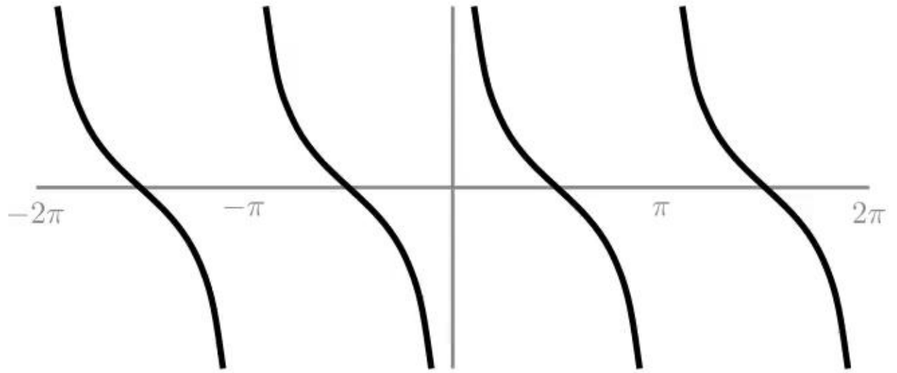


D)

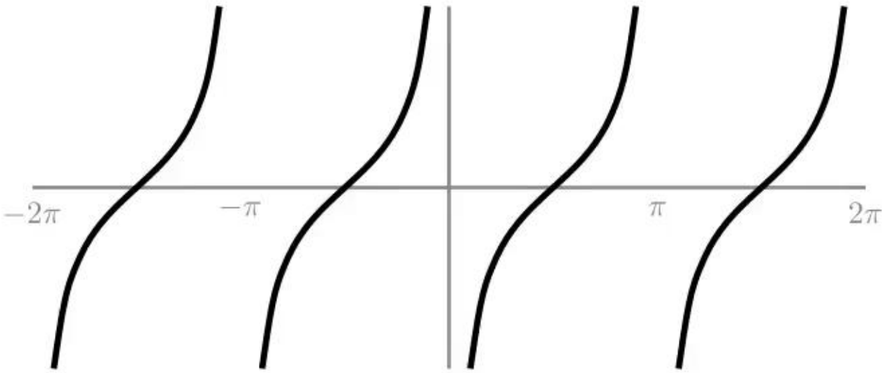


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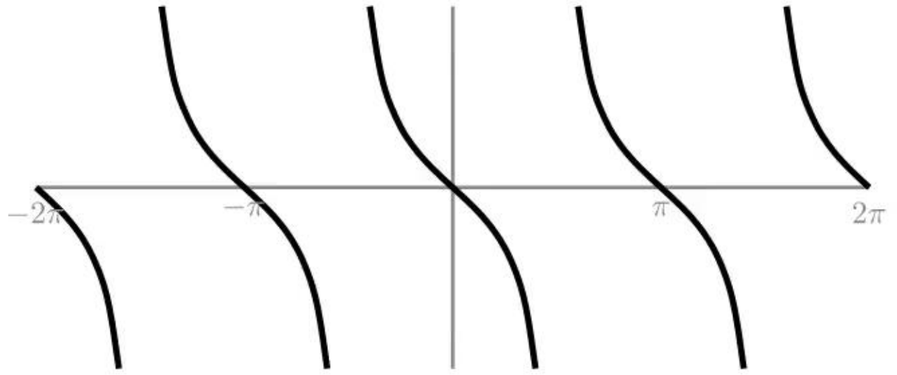
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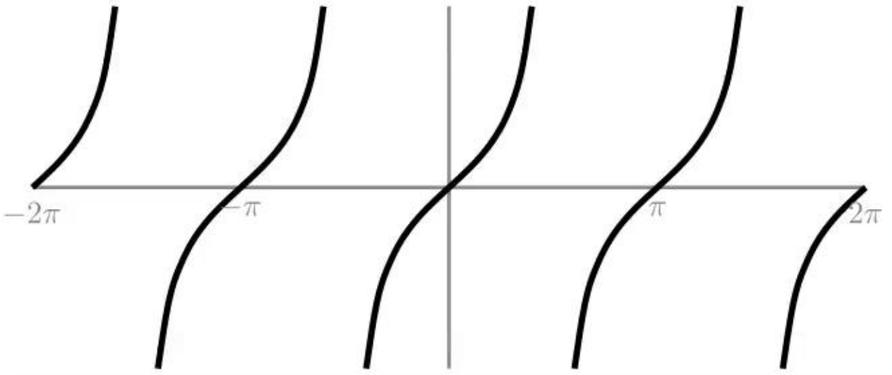
B)



C)



D)



Tip 1: Try some *convenient* numbers

$$\sin\left(x + \frac{\pi}{2}\right) = ?$$

A) $\cos(x)$ B) $-\cos(x)$

C) $\sin(x)$ D) $-\sin(x)$

$$\sin\left(x+\frac{\pi}{2}\right)=?$$

A) $\cos(x)$

B) $-\cos(x)$

C) $\sin(x)$

D) $-\sin(x)$

The derivative of $\sin(x)$ is:

A) $\cos(x)$

B) $-\cos(x)$

C) $\sin(x)$

D) $-\sin(x)$

The derivative of $\sin(x)$ is:

A) $\cos(x)$

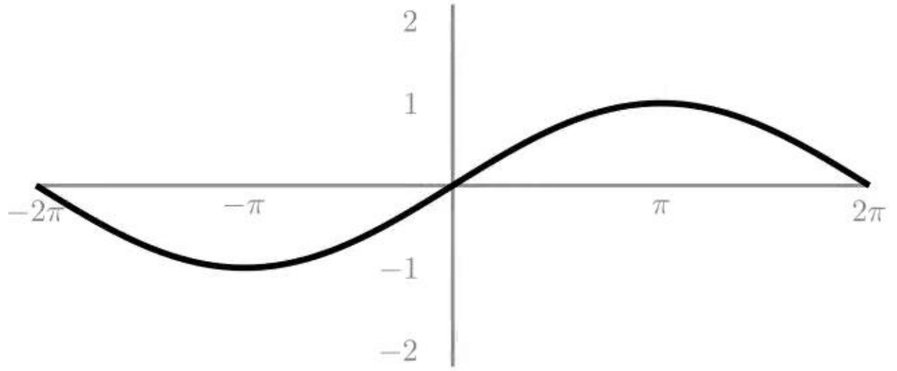
B) $-\cos(x)$

C) $\sin(x)$

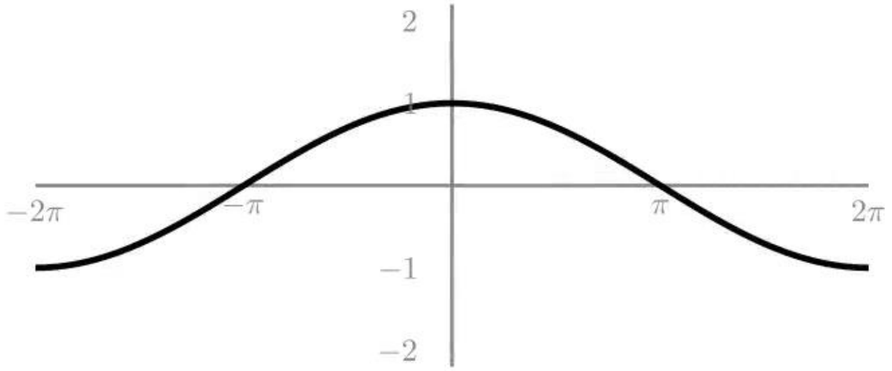
D) $-\sin(x)$

Which of the following graphs is $y = \sin(2x)$?

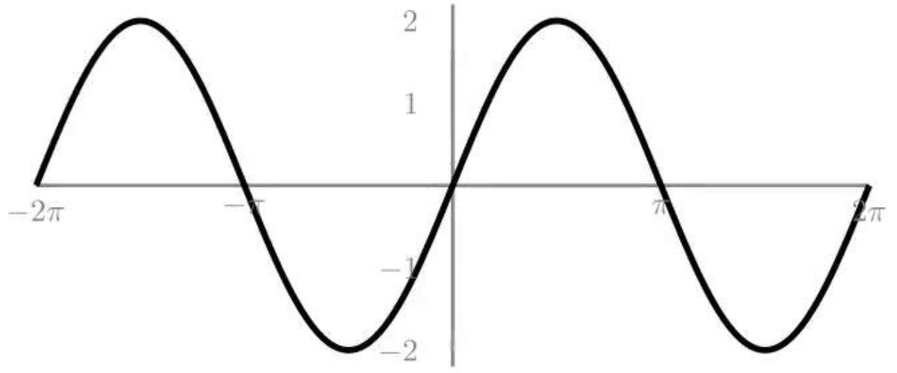
A)



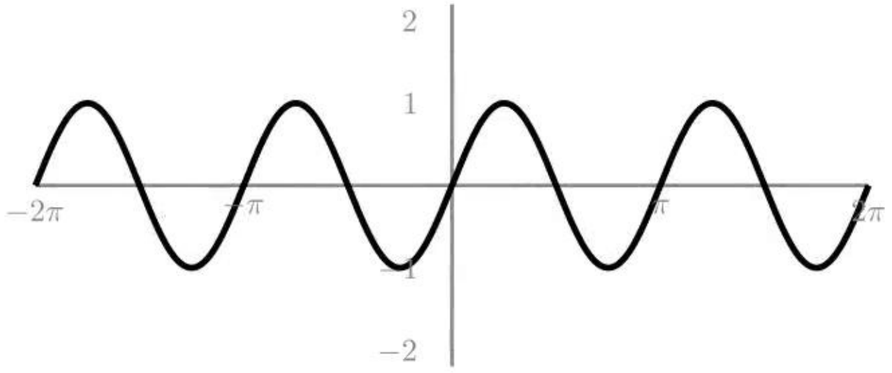
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C)

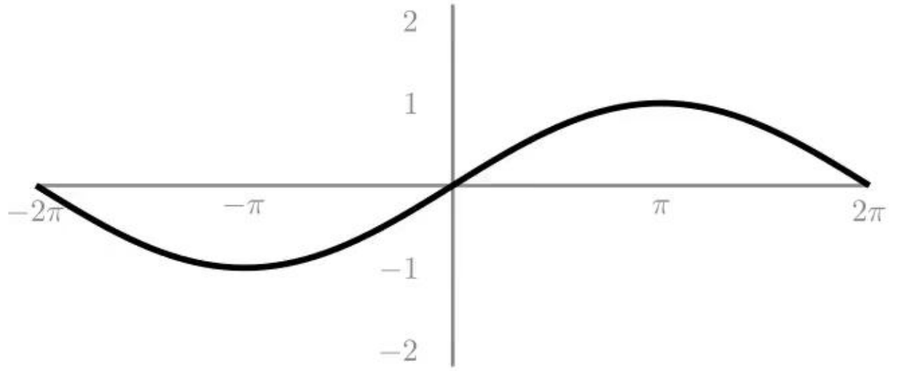


D)

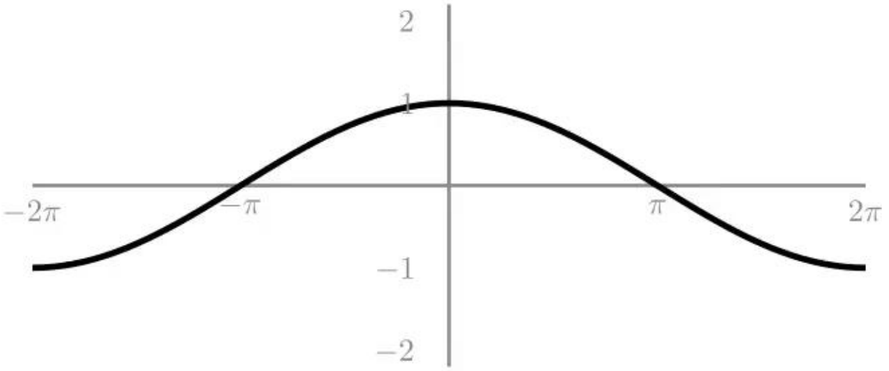


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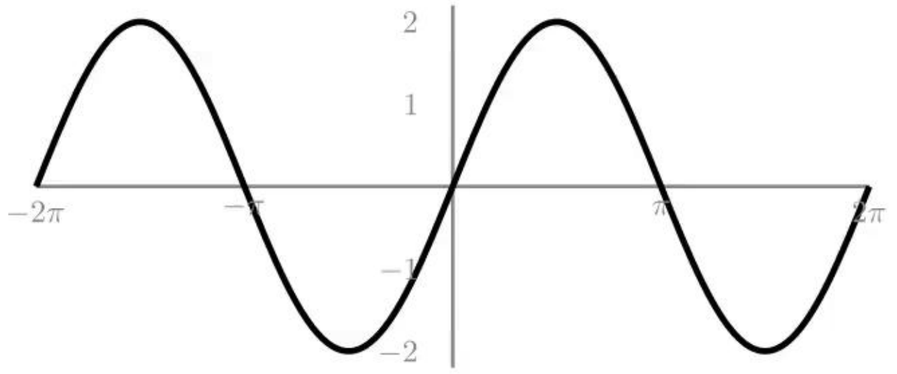
A)



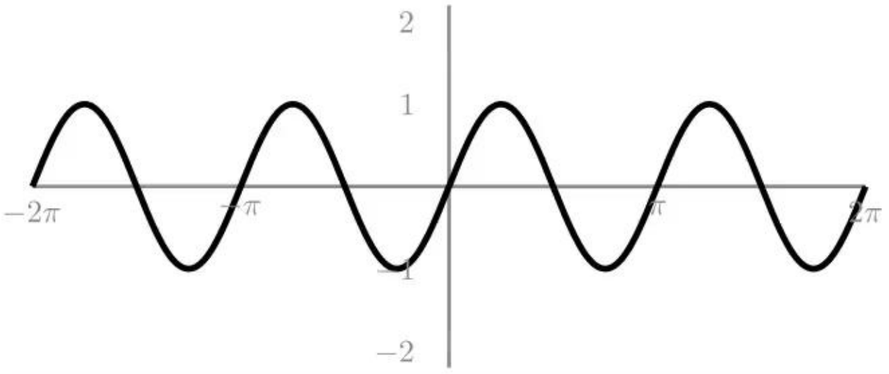
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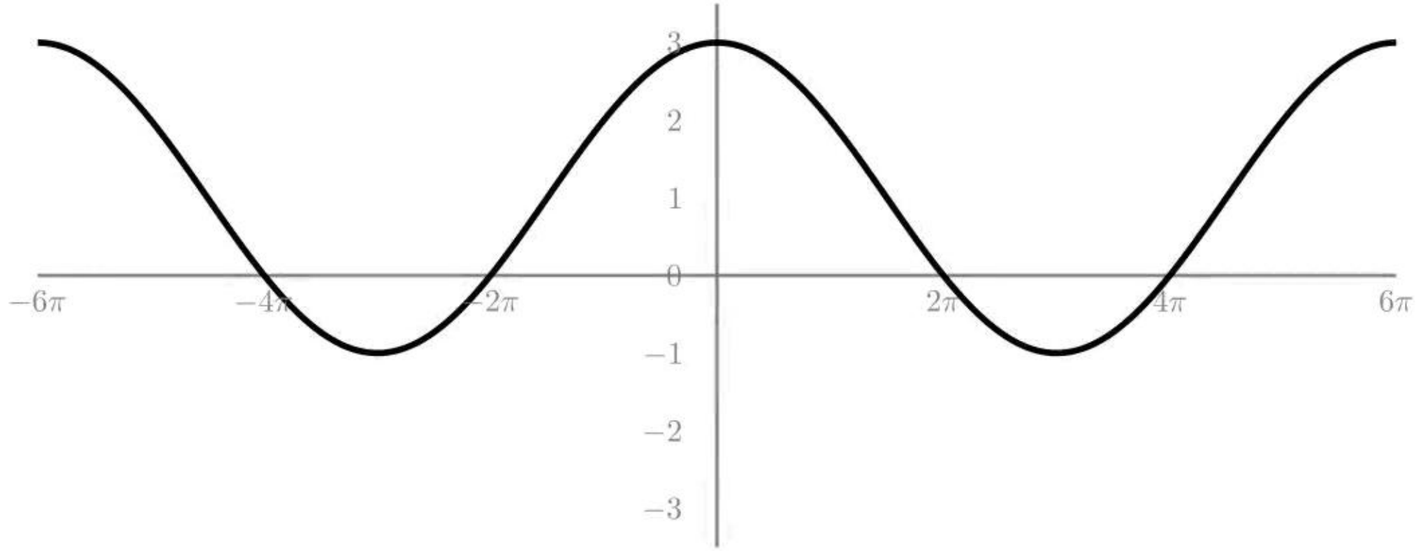


C)



D)





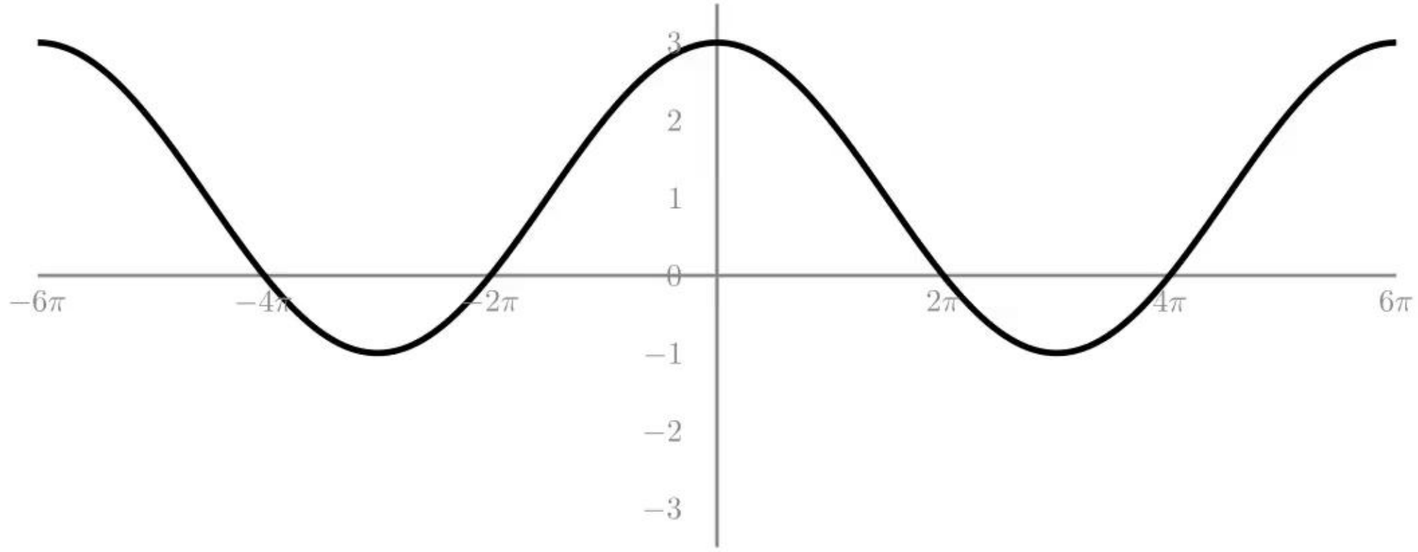
What is the function corresponding to this graph?

A) $1 + 2\cos\left(\frac{x}{3}\right)$

B) $3 - 2\sin(x)$

C) $2 + \cos(3x)$

D) $1 + 2\cos(3x)$



What is the function corresponding to this graph?

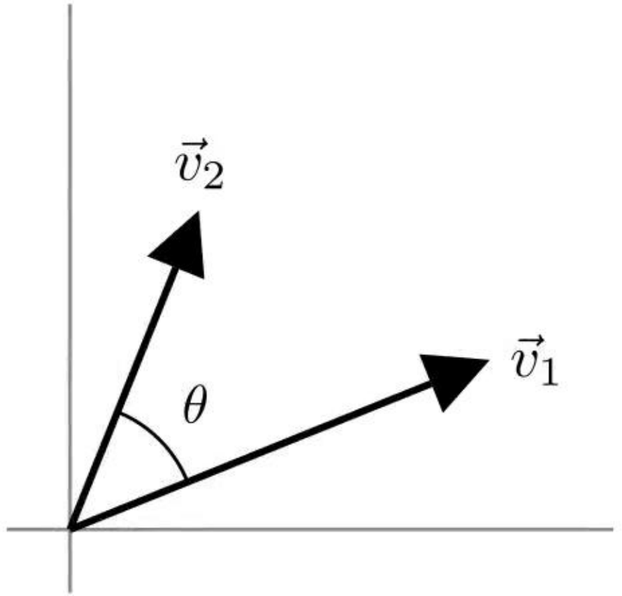
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C) $2 + \cos(3x)$

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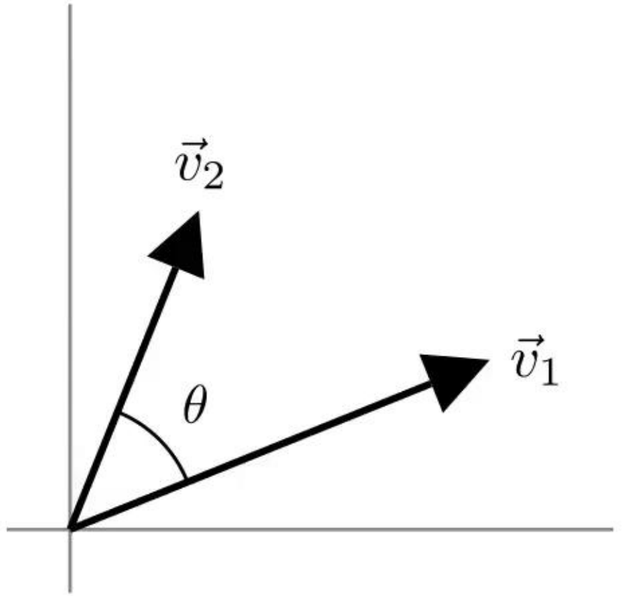
The area of the parallelogram is found by the:



C) Cross product

D) Dot product

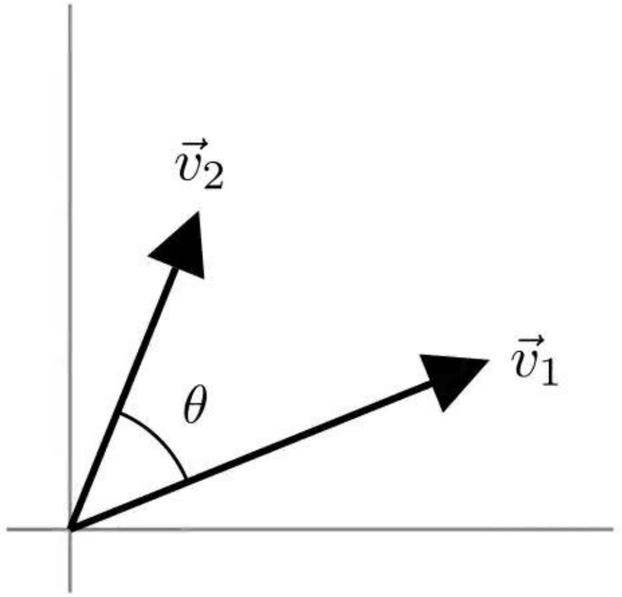
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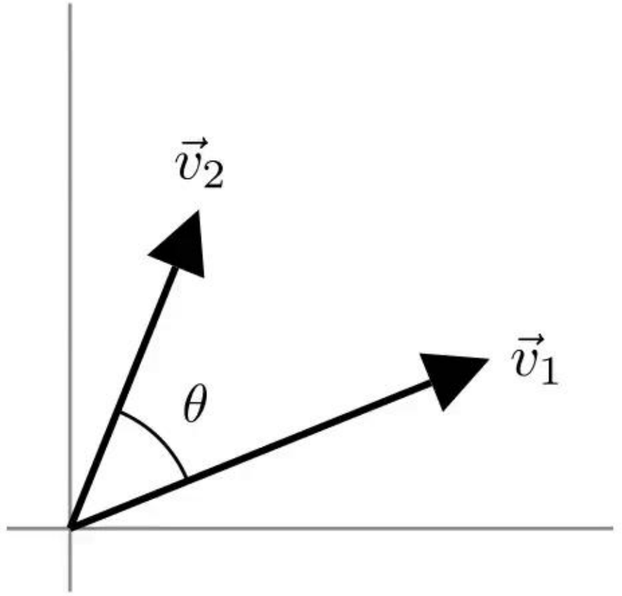
The length of the cross product $|\vec{v}_1 \times \vec{v}_2|$ is:



C) $|\vec{v}_1|\,|\vec{v}_2| \cos\theta$

D) $|\vec{v}_1|\,|\vec{v}_2| \sin\theta$

The length of the cross product $|\vec{v}_1 \times \vec{v}_2|$ is:



C) $|\vec{v}_1|\,|\vec{v}_2| \cos\theta$

D) $|\vec{v}_1|\,|\vec{v}_2| \sin\theta$

If $T = \tan(x)$ and $0 \leq x \leq \dfrac{\pi}{2}$, what is $\sin(x)$?

A) $\dfrac{T}{\sqrt{1+T^2}}$

B) $\dfrac{1}{\sqrt{1+T^2}}$

C) $\dfrac{\sqrt{1-T^2}}{T}$

D) $\sqrt{1-T^2}$

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A) $\dfrac{T}{\sqrt{1+T^2}}$

B) $\dfrac{1}{\sqrt{1+T^2}}$

C) $\dfrac{\sqrt{1-T^2}}{T}$

D) $\sqrt{1-T^2}$

Tip 1: Try some *convenient* numbers

Tip 2: Draw a picture

2D Geometry and Parametric Equations

The point $(1, 2)$ is rotated $90°$ clockwise about the origin. Where does it land?

A) $(2, 1)$

B) $(1, -2)$

C) $(-2, 1)$

D) $(2, -1)$

The point $(1, 2)$ is rotated $90°$ clockwise about the origin. Where does it land?

A) $(2, 1)$

B) $(1, -2)$

C) $(-2, 1)$

D) $(2, -1)$

Tip 1: Try some *convenient* numbers

Tip 2: Draw a picture

Tip 3: Coherence check (back-substitution)

extra stuff: vectors and sin addition

The graph of $y = e^x$ is reflected across the line $y = x$. What do you get?

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B) $-e^x$

C) $\log(x)$

D) not a function

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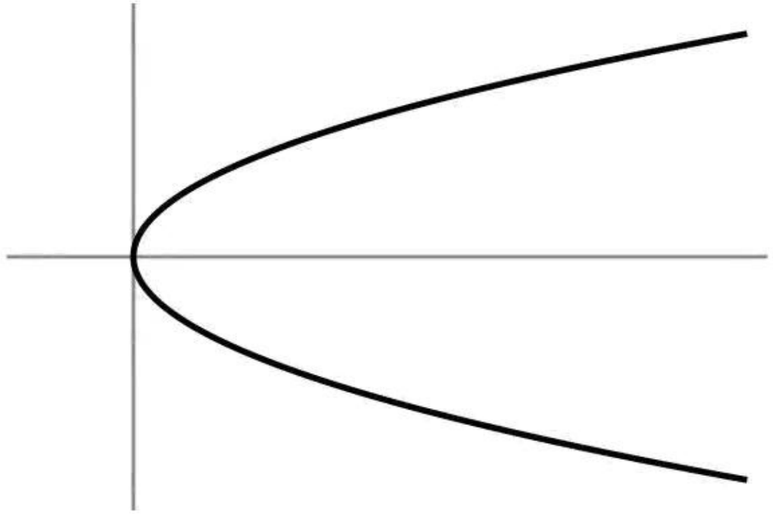
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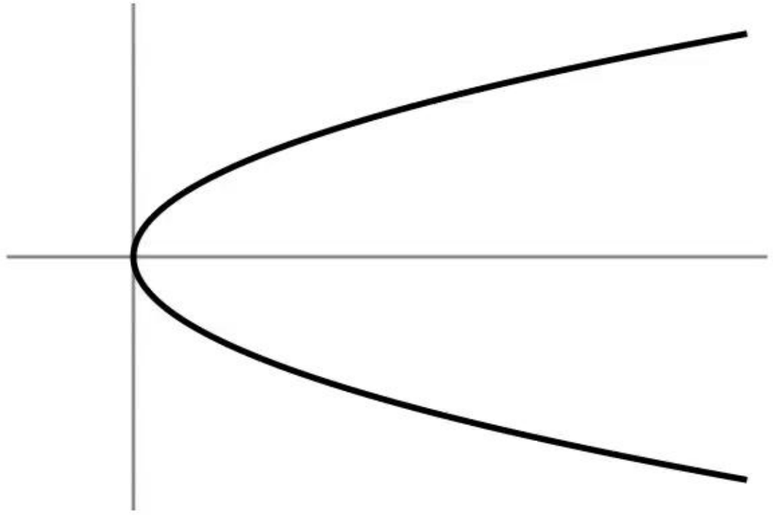
A) y is a function of x AND x is a function of y

B) y is a function of x AND x is NOT a function of y

C) y is NOT a function of x AND x is a function of y

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Same question for $x^2 + y^2 = 1$. Which is true?

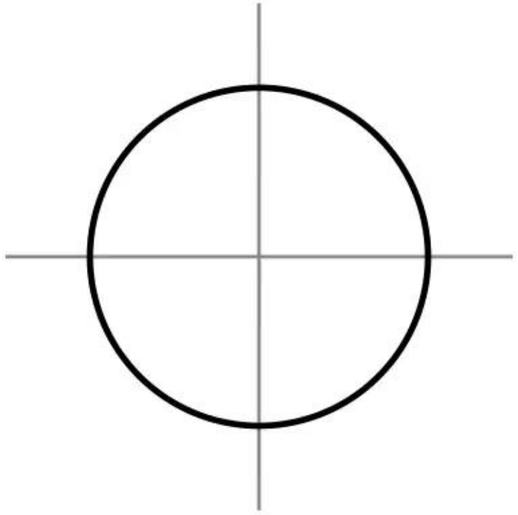
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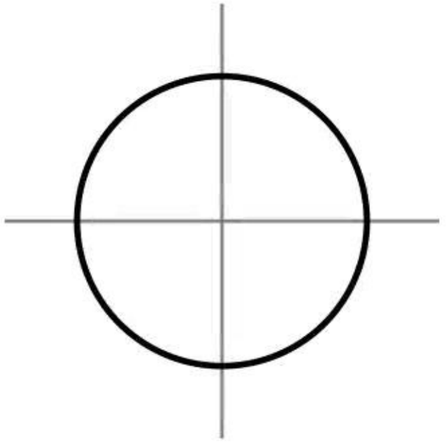
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Which of the following is NOT a parametrization of $x^2 + y^2 = 1$?



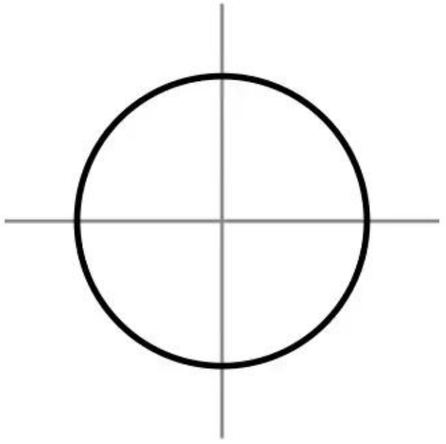
A) $(\sin t,\ \cos t), \quad 0 \le t \le 2\pi$

B) $(\cos(2\pi \sin t),\ \sin(2\pi \sin t)), \quad 0 \le t \le \frac{\pi}{2}$

C) $\left(\cos \frac{t}{2},\ \sin \frac{t}{2}\right), \quad 0 \le t \le 2\pi$

D) $(\sin t,\ \cos t), \quad 0 \le t \le 4\pi$

Which of the following is NOT a parametrization of $x^2 + y^2 = 1$?



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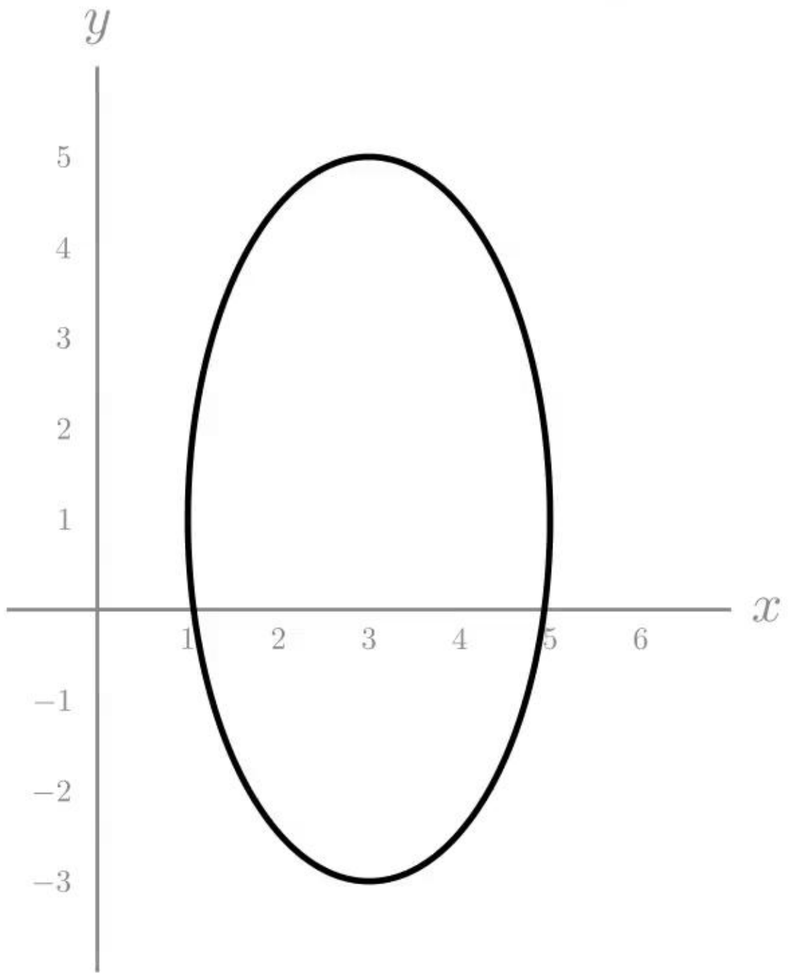
D) $(\sin t,\ \cos t), \quad 0 \le t \le 4\pi$

Draw the curve parametrized by:

$$x = 3 - 2\cos(t) \qquad y = 1 + 4\sin(t), \qquad 0 \leq t \leq 2\pi$$

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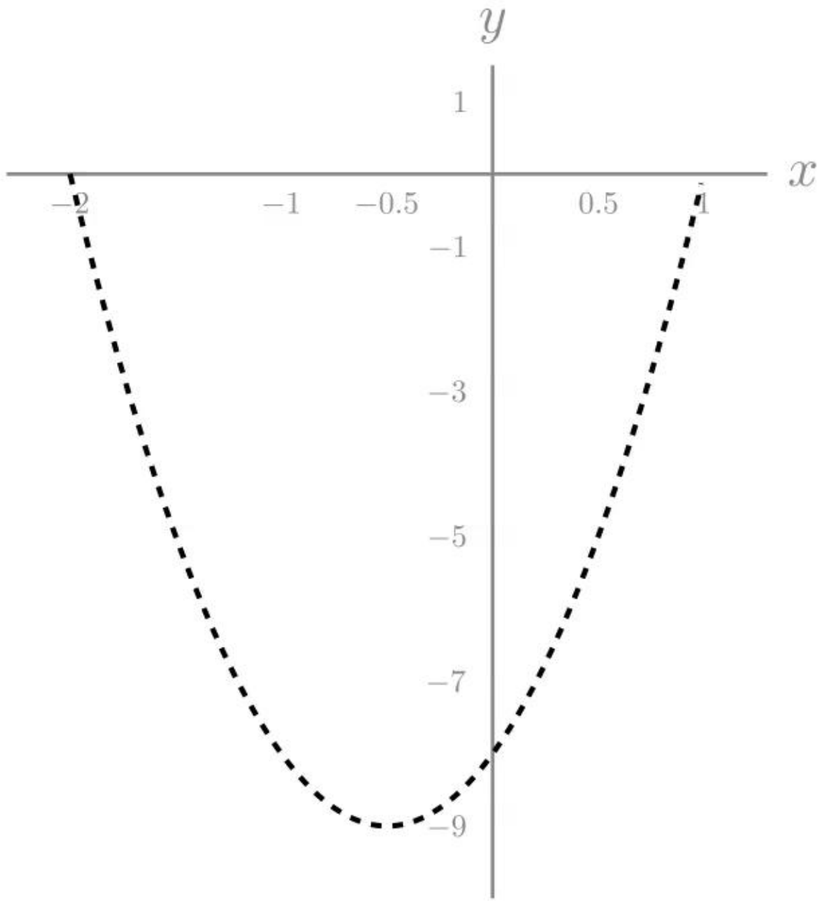


Draw the curve parametrized by:

$$x = \tfrac{1}{2}e^{-3t} \qquad y = e^{-6t} + 2e^{-3t} - 8$$

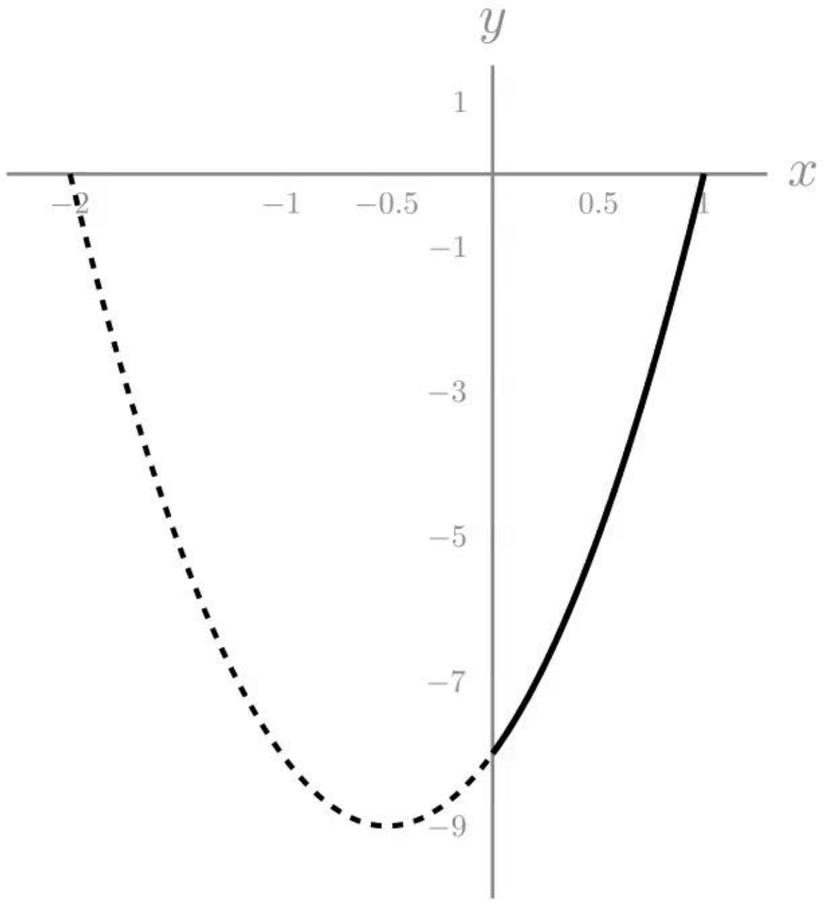
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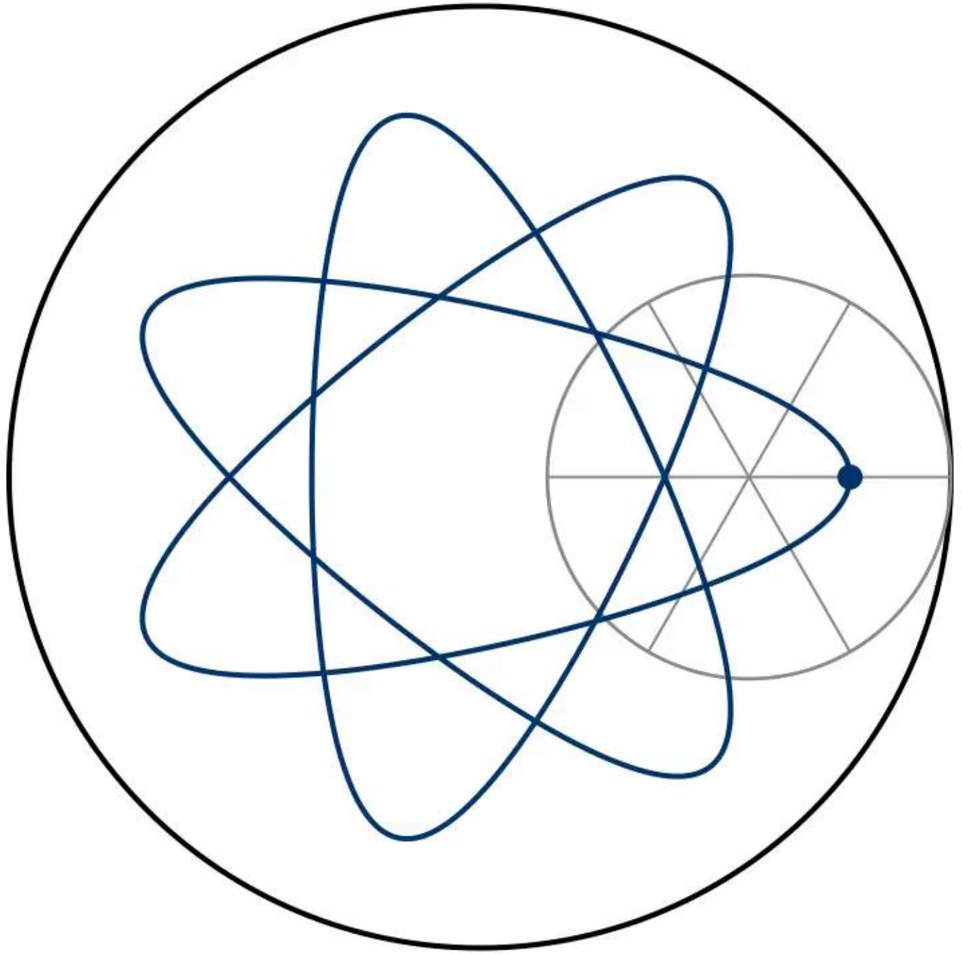


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spirograph



Hidden Curriculum

- Study strategies
- Student struggles
- (s)Tech support